

Birth-death models for estimating speciation and epidemiological parameters

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Taming the BEAST – Beijing, July 2025



Birth-death models

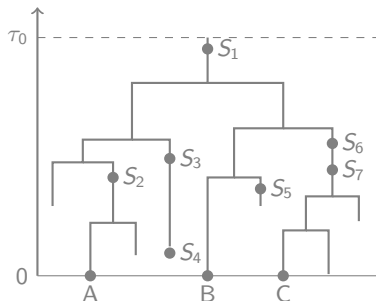
Identifiability of birth-death model parameters

Stratigraphic-range birth-death model

Birth-death models

Constant rate birth-death sampling (BDS) model

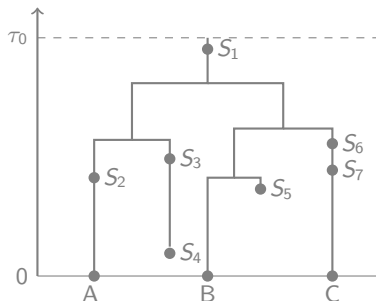
The process starts at time of origin $\tau_0 > 0$ and ends at time zero (present time).



- ▶ λ – birth rate
- ▶ μ – death rate ($\lambda \geq \mu$)
- ▶ ψ – sampling rate
- ▶ r – removal probability
- ▶ ρ_0 – sampling at present probability

Sampled or reconstructed tree

The process starts at time of origin $\tau_0 > 0$ and ends at time zero (present time).

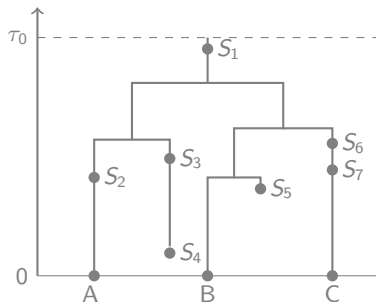


- ▶ λ – birth rate
- ▶ μ – death rate
- ▶ ψ – sampling rate
- ▶ r – removal probability
- ▶ ρ_0 – sampling at present probability

The probability density function, $P(T|\lambda, \mu, \psi, r, \rho_0, \tau_0)$, has an analytical form.

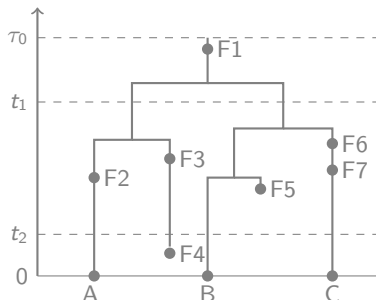
Time dependant rates

The process starts at time of origin $\tau_0 > 0$ and ends at time zero (present time).



- ▶ $\lambda(t)$ – birth rate
- ▶ $\mu(t)$ – death rate
- ▶ $\psi(t)$ – sampling rate
- ▶ $r(t)$ – removal probability
- ▶ ρ_0 – sampling at present probability

The process starts at time of origin $\tau_0 > 0$ and ends at time zero (present time).

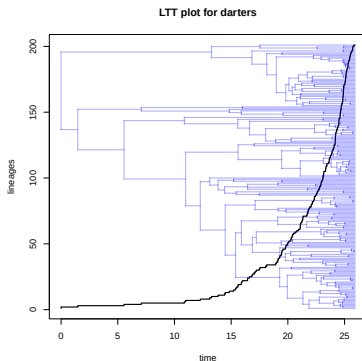


- ▶ $\lambda_1, \dots, \lambda_k$ – birth rates
- ▶ $\mu_1, \dots, \mu_k$ – death rates
- ▶ $\psi_1, \dots, \psi_k$ – sampling rates
- ▶ $r_1, \dots, r_k$ – removal probabilities
- ▶ ρ_0 – sampling at present probability

The probability density function, $P(\mathcal{T} | \bar{\lambda}, \bar{\mu}, \bar{\psi}, \bar{r}, \rho_0, \tau_0)$, has an analytical form.

Identifiability of birth-death model parameters

Dated phylogenetic tree or LTT for parameter inference



Question:

Is a dated phylogenetic tree or LTT plot enough to infer speciation and extinction rates?

A statistical model $\mathcal{P}_\Theta = \{P_\theta : \theta \in \Theta\}$ is **identifiable** if for all $\theta_1, \theta_2 \in \Theta$:

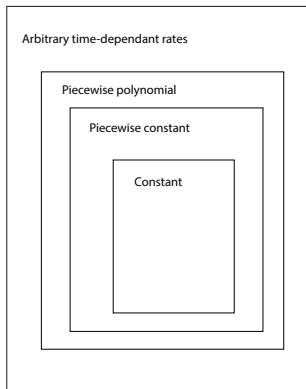
$$P_{\theta_1} \stackrel{a.e.}{=} P_{\theta_2} \text{ implies } \theta_1 \stackrel{a.e.}{=} \theta_2.$$

In the case of birth-death models:

- ▶ $P_\theta = P(\mathcal{T}|\lambda, \mu, \psi, r, \rho_0, \tau_0)$
- ▶ For identifiability results we always assume τ_0 is fixed: $\theta = (\lambda, \mu, \psi, r, \rho_0)$
- ▶ We also consider $P(\mathcal{T}|\theta, S)$ the distribution conditional on sampling of at least one individual either via ψ or ρ_0 sampling

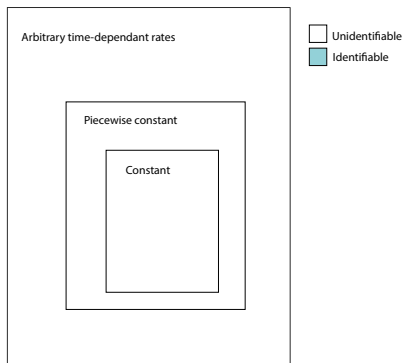
Parameter classes

- ▶ Highest class for identifiability
- ▶ Smallest class for unidentifiability



Birth-death model with present sampling

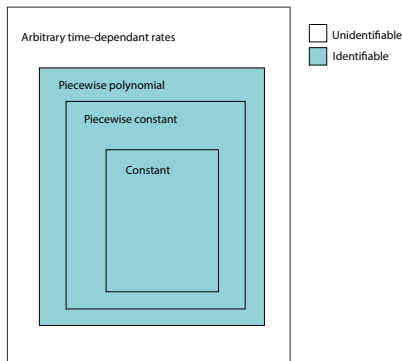
$$\lambda(t), \mu(t), \rho_0$$



- ▶ Stadler 2010: constant parameters
- ▶ All unidentifiability results here are for model conditioning on sampling of at least one individual

Birth-death model with present sampling

$\lambda(t)$, $\mu(t)$, and ρ_0 - fixed to a value in $(0, 1]$

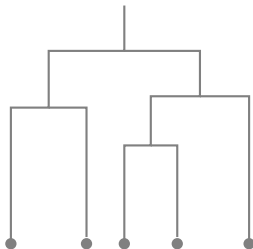


- ▶ Legried and Terhorst 2022, 2023: piecewise constant and piecewise polynomial parameters
- ▶ Kubo & Iwasa 1995, Louca & Pennell 2020: general time-dependant parameters

Equivalent parameterisation

- ▶ Let $E(t)$ be the probability of a lineage alive at time t to not leave sampled descendants
- ▶ Time-dependant BD model is uniquely defined by:

$$\tilde{\lambda} = \lambda(1 - E) - \text{pulled birth rate}$$

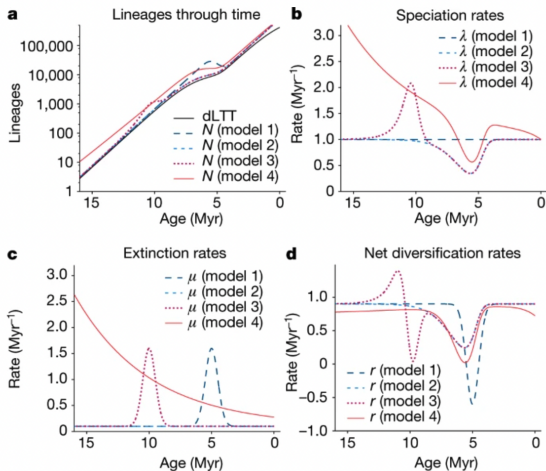


Reconstructed tree

- ▶ **dLTT** is the **expected** number of lineages through time in the **reconstructed** tree under the BD model
- ▶ Two models that have the same dLTT have the same probability densities over the space of trees

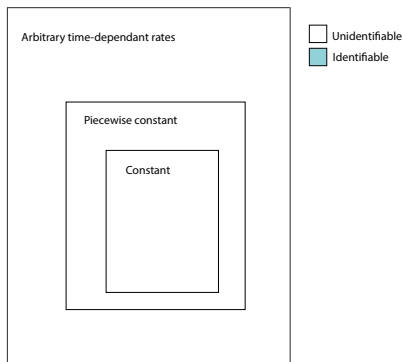
Time-dependant BD model congruence classes (Louca & Pennell 2020)

Fig. 1: Illustration of congruent birth–death processes (simulations).



Birth-death-sampling model

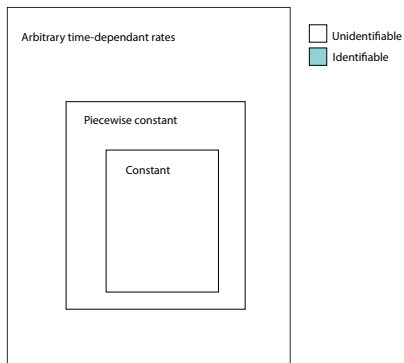
$$\lambda(t), \mu(t), \psi(t), r(t), \rho_0$$



- ▶ Stadler *et al.* 2013, Gavryushkina *et al.* 2014: constant and piecewise constant parameters
- ▶ Truman *et al.* 2025: arbitrary time-dependant parameters

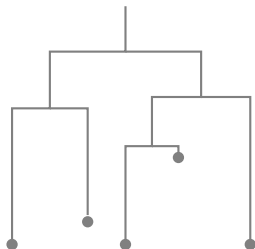
Birth-death-sampling model $r = 1, \rho_0 = 0$

$\lambda(t), \mu(t), \psi(t)$, with $r = 1$ and $\rho_0 = 0$



- ▶ Stadler 2013: constant rates
- ▶ Louca *et al.* 2021: time-dependant rates

Time-dependant BDS model conditioned on sampling is uniquely defined by



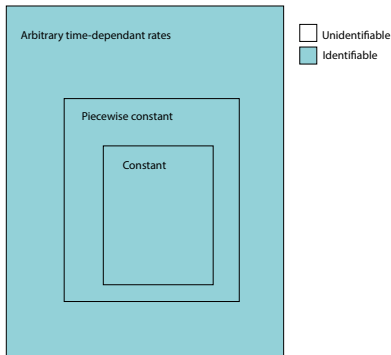
$$\tilde{\lambda} = \lambda(1 - E) - \text{pulled birth rate}$$

$$\tilde{\psi} = \frac{\psi}{1 - E} - \text{pulled sampling rate}$$

For arbitrary time-dependant rates an analytical solution for $E(t)$ is not known.

Birth-death-sampling model r is fixed

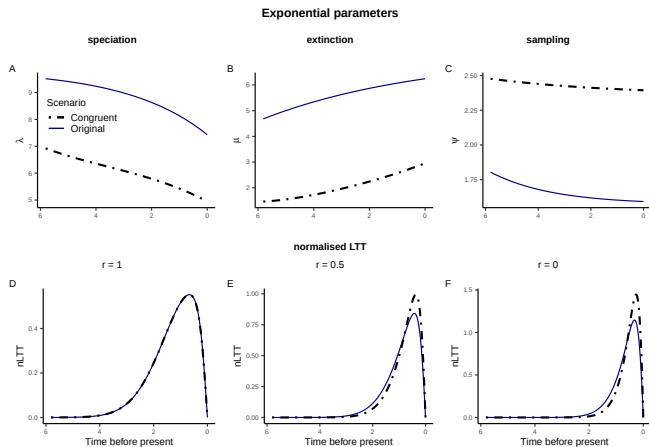
$\lambda(t), \mu(t), \psi(t), r(t)$ is fixed to a function that take values in $[0, 1), \rho_0$



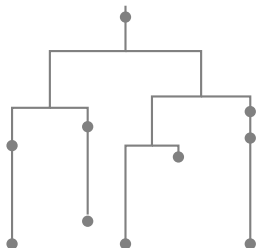
This class includes FBD model, when $r(t) = 0$.

- ▶ Truman et al 2025: arbitrary time-dependant rates

Congruence classes break for $r = 0$ or $r = 0.5$



The distribution of reconstructed trees under a time-dependant BDS model with $\psi(t) > 0$ conditioned on sampling is uniquely defined by



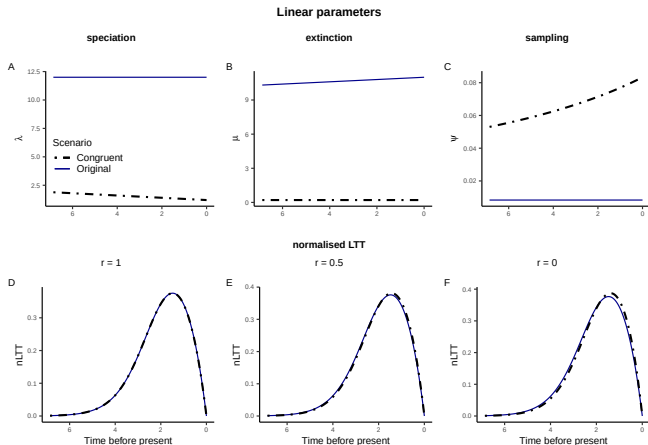
$$\tilde{\lambda} = \lambda(1 - E)$$

$$\tilde{\mu} = \frac{\psi E}{1 - E} + \psi r$$

$$\tilde{\psi} = \psi(1 - r)$$

For arbitrary time-dependant rates an analytical solution for $E(t)$ is not known.

Small sampling rate ψ



- If the sampling rate is small the differences in the distributions might be very small

- ▶ In practice we restrict the class of rates in the inference, e.g., to constant or piece-wise constant
- ▶ If the true rates are piece-wise constant then identifiable models would estimate true parameters
- ▶ How close the piecewise constant estimates of the parameters to the true non-piecewise constant parameters?
- ▶ The precision of the parameter estimates depends on the number of birth, death, and sampling events within the intervals
- ▶ If the model is unidentifiable, fixing some parameters or using strong priors could help

Stratigraphic-range birth-death model

Total-evidence dating using FBD ($r = 0$)



Fossil ages, τ (or age intervals)



Morphological data of extinct and extant species, D_{morph}



Molecular data of extant species, D_{mol}

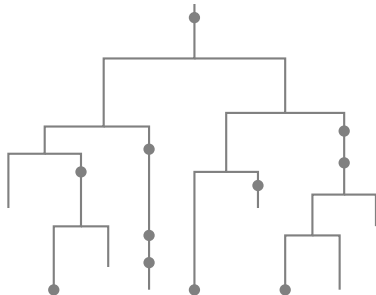
Denote $D = (D_{morph}, D_{mol})$

Using Bayesian inference we can estimate the joint posterior distribution of

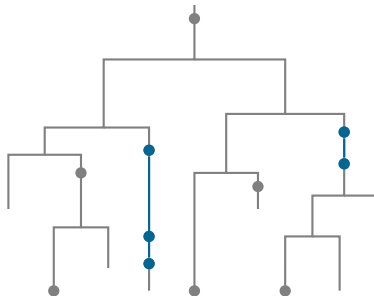
- ▶ Dated phylogenetic tree, $\mathcal{T}$
- ▶ Diversification parameters, θ
- ▶ Substitution model parameters, η

$$P(\mathcal{T}, \theta, \eta | D, \tau) \propto P(D | \mathcal{T}, \eta) P(\mathcal{T} | \theta, \tau) P(\theta, \eta)$$

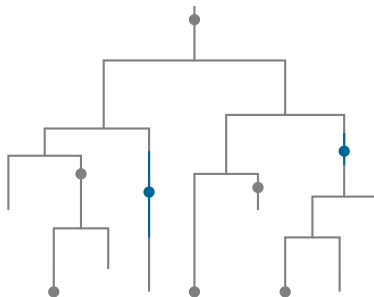
Modelling fossil sampling in FBD



- Several fossil samples along a lineage can belong to the same species

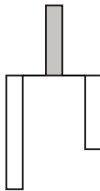
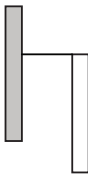


- Such cases were modelled as a single occurrence with age boundaries spanning the stratigraphic range of the species



Three types of speciation (Foote 1996)

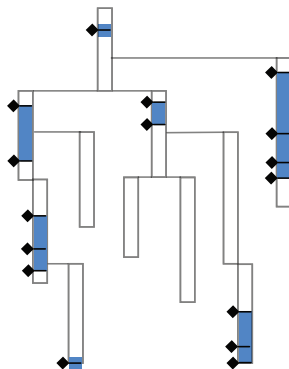
(i) asymmetric speciation (ii) symmetric speciation (iii) anagenetic speciation



Mixed speciation stratigraphic range fossilised birth-death process

Stadler *et al.* 2017

- ▶ Branching rate, λ ,
- ▶ Death rate, μ ,
- ▶ Fossil sampling rate, ψ ,
- ▶ Probability of a symmetric branching event, β ,
- ▶ Anagenetic speciation rate, λ_a , and
- ▶ Probability of sampling at present, ρ .



Silvestro *et al.* 2018

Phylogenetic approach:

speciation rate λ

extinction rate μ

Fossil stratigraphic range approach:

$$\lambda^* = \lambda(1 - \beta) + 2\lambda\beta + \lambda_a \geq \lambda$$

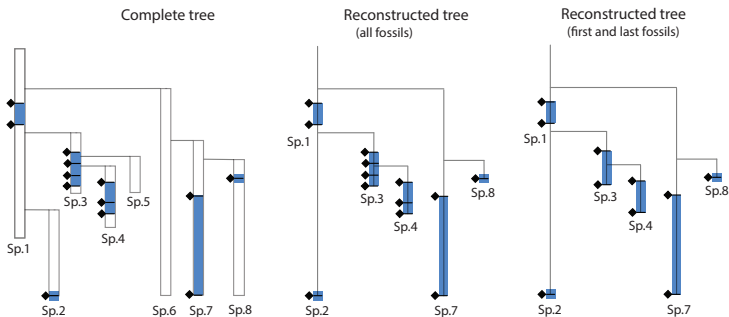
$$\mu^* = \lambda\beta + \lambda_a + \mu \geq \mu$$

$$\begin{array}{l} \lambda = \lambda^* \\ \mu = \mu^* \end{array} \quad \text{only if} \quad \beta = \lambda_a = 0$$

Net diversification rates are equal:

$$d = \lambda - \mu = \lambda^* - \mu^*$$

SRFBD: pure budding process $\beta = \lambda_a = 0$



The probability density of an oriented reconstructed tree with intermediate fossils removed has an analytical expression.

Total evidence dating with SRFBD

Using [sRanges](#) (Stolz, Gavryushkina *et al.* 2025) package for BEAST2, we can estimate SR oriented trees and diversification parameters in total evidence dating:



Fossil stratigraphic ranges represented by the ages (or intervals) of the first and the last occurrences, τ



Morphological data of extinct and extant species, D_{morph} (placed on one or both ends of SR)



Molecular data of extant species, D_{mol}

Total evidence FBD (Gavryushkina *et al.* 2017):

- ▶ Morphological matrix of 36 extinct and 19 extant species,
- ▶ Molecular data of extant species, and
- ▶ Fossil ages with uncertainty intervals
- ▶ Lewis Mk model for morphological evolution
- ▶ Independent relaxed clock models for molecular and morphological data
- ▶ FBD model with ρ fixed to one.

Total-evidence SRFBD (Stolz, Gavryushkina *et al.* 2025)

- ▶ FBD: with updated ages
- ▶ SRFBD with morphological data placed on the first occurrence
- ▶ SRFBD with morphological data placed on both, the first and the last occurrence

Penguins analysis: estimated parameters

Diversification rate

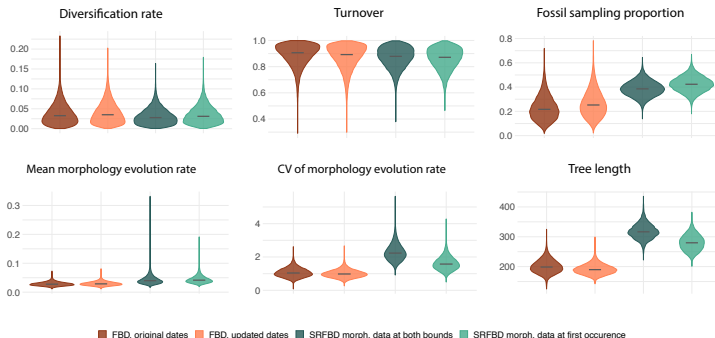
$$d = \lambda - \mu$$

Turnover

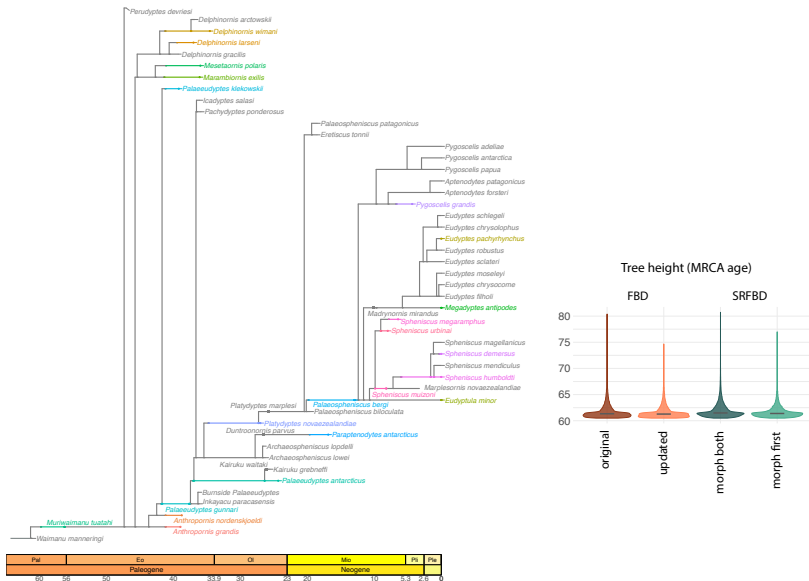
$$\nu = \frac{\mu}{\lambda}$$

Sampling proportion

$$s = \frac{\psi}{\psi + \mu}$$



Penguins analysis: random tree from SRFBD posterior distribution



Analysis of North American canids

We reanalysed morphological data of five extant and 116 extinct North American canids from Slater 2015.

- ▶ FBD

- ▶ ages are fixed as in Slater 2015
 - ▶ ρ is estimated

- ▶ FBD

- ▶ updated ages with uncertainty intervals
 - ▶ $\rho = 0$, because all extant species also have fossil samples with the present time as the lower age boundary

- ▶ SRFBD

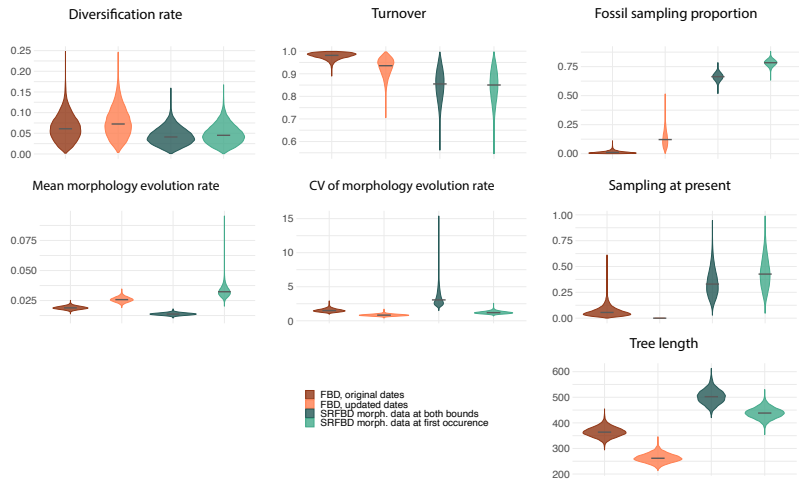
- ▶ first and last occurrences with uncertainty intervals
 - ▶ ρ is estimated
 - ▶ morphological data on the first occurrence

- ▶ SRFBD

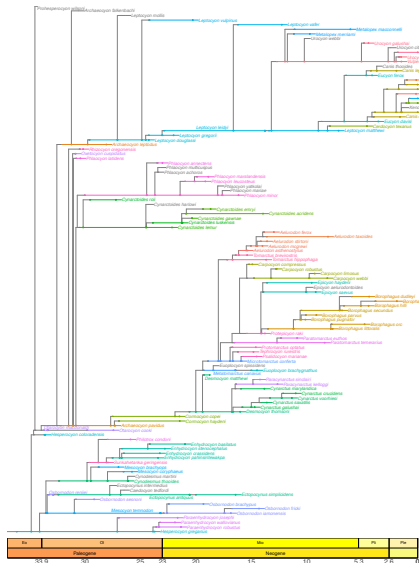
- ▶ first and last occurrences with uncertainty intervals
 - ▶ ρ is estimated
 - ▶ morphological data on both ends



Canids analysis: estimated parameters

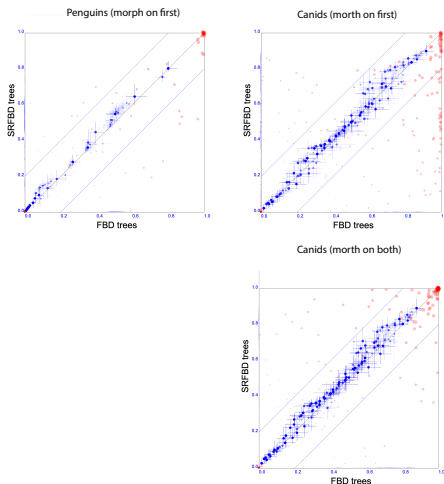


Canids analysis: random tree from SRFBD posterior distribution



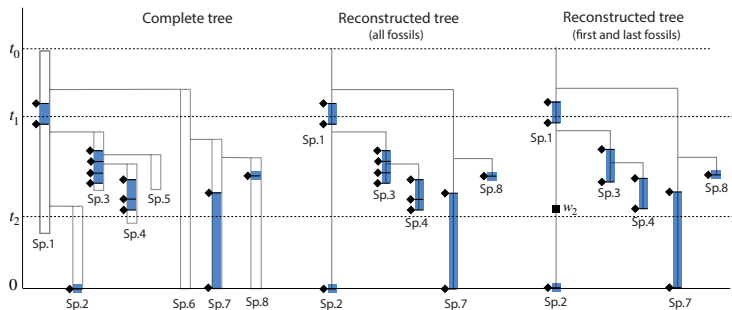
Clade comparison for penguins and canids

- In both analyses, the estimated ages in SRFBD are relatively older than in FBD
- For canids, placement of the morphological data on both ends makes the distribution of clades more similar to FBD



Piecewise constant (skyline) SRFBD

Truman *et al.* in progress

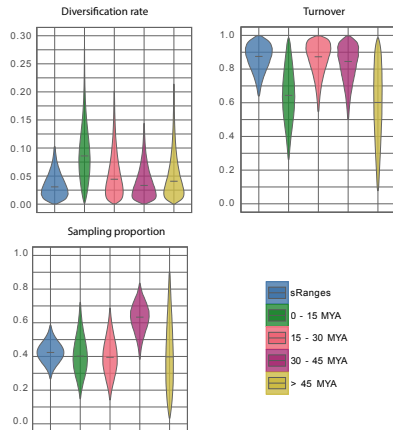


The likelihood function for a SR oriented tree with the first and last occurrences has an analytical expression:

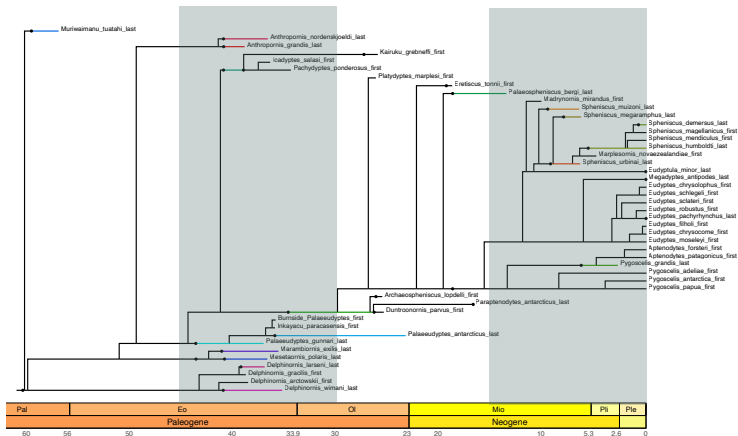
$$\begin{aligned}
 f[\mathcal{T}_{s,r}^o | \bar{\lambda}, \bar{\mu}, \bar{\psi}, \rho, \bar{t}] &= \rho^{\sum_{i=1}^n \chi_i} \prod_{\gamma=1}^g \lambda_{\omega(x_\gamma)}(x_\gamma) \prod_{i=1}^n [1 + (1 - h_i)(1 - \chi_i)(p_{\omega(y_i)}(y_i) - 1)] \prod_{i=1}^n \psi_{\omega(o_i)} t_i \\
 &\times e^{\sum_{i=1}^n \delta \mathcal{R}_i \psi_{\omega(o_i)}(o_i - y_i)} \times e^{\sum_{i=1}^n [(1 - \delta \mathcal{R}_i)(\psi_{\omega(o_i)}(o_i - t_{\omega(o_i)}) + \psi_{\omega(y_i)}(t_{\omega(y_i)} - 1 - y_i)) + \sum_{j=\omega(o_i)+1}^{\omega(y_i)-1} (\psi_j(t_{j-1} - t_j))]} \\
 &\times \prod_{i=1}^f \hat{q}_{\omega(B_i)}(B_i) \prod_{i \in I_u} \left(1 - \frac{q_{\omega(o_i)}(o_i) \tilde{q}_{\omega(y_{a(i)})}(y_{a(i)})}{\tilde{q}_{\omega(o_i)}(o_i) q_{\omega(y_{a(i)})}(y_{a(i)})}\right) \\
 &\times \prod_{i \in I \setminus I_u} \left[\left(1 - \frac{q_{\omega(o_i)}(o_i) \tilde{q}(t_{\omega(o_i)} - 1)}{\tilde{q}_{\omega(o_i)}(o_i) q(t_{\omega(o_i)} - 1)}\right) + \left(\frac{q_{\omega(o_i)}(o_i)}{\tilde{q}_{\omega(o_i)}(o_i)} \left(1 - \frac{\tilde{q}(y_{a(i)})}{q(y_{a(i)})}\right) \prod_{k=\omega(y_{a(i)})}^{\omega(o_i)-1} \frac{\tilde{q}_{k+1}(t_k)}{q_{k+1}(t_k)}\right) \right. \\
 &\left. + \frac{q_{\omega(o_i)}(o_i)}{\tilde{q}_{\omega(o_i)}(o_i)} \sum_{e=\omega(o_i)-1}^{\omega(y_{a(i)})+1} \left((1 - \frac{\tilde{q}_e(t_{e-1})}{q_e(t_{e-1})}) \prod_{k=e}^{\omega(o_i)-1} \frac{\tilde{q}_{k+1}(t_k)}{q_{k+1}(t_k)}\right) \right]
 \end{aligned}$$

Skyline SRFBD penguins analysis

- ▶ The time is split in four ~ 15 Myr long intervals
- ▶ Morphological data is attached to the first occurrences

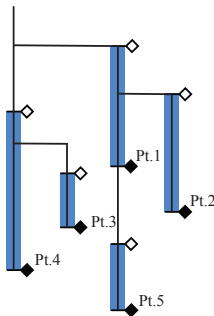


Penguins skyline SRFBD analysis



SRFBD epidemiological applications

- ▶ If multiple samples per patient are available one can use SRFBD directly (Stolz *et al.* 2024)
- ▶ When transmission time is known (e.g. HIV), we can extend this model to infection interval BD model



Assuming that all individuals survive until present, we can derive the exact likelihood

Identifiability project:

- ▶ Kate Truman
- ▶ Tim Vaughan
- ▶ Alex Gavryushkin

sRanges project:

- ▶ Ugne Stolz
- ▶ Bethany Allen
- ▶ Tim Vaughan
- ▶ Tanja Stadler
- ▶ Kate Truman (skyline SRFBD)
- ▶ Denise Kuhnert (infection intervals)

Thank you!